

NYC Taxi Trip Duration — Regression Report

23CSE301 Machine Learning | Review 1 | Regression track

Data file: /Users/bettim/Documents/Semester 5/Machine Learning/Capstone Evaluation/data/nyc_taxi_trip_duration.csv

Raw rows loaded: 352,000

Rows after cleaning: 349,080

Rows in modelling table: 349,080

Training rows: 279,264 Evaluation rows: 50,000

FAST_DEV = False

Best model by test R^2 : Gradient Boosting Regressor

Official models trained: 12

Figures exported: 22

Pipeline: audit → EDA → clean → engineer → split/encode/scale

→ 10 models → compare/tune/CV → manual tester.

Encoders and scalers were fit on the training split only.

dropoff_datetime and id were never used as predictors.

Data cleaning — rows removed at each step

step	rows_before	rows_after	rows_removed	pct_removed	note
loaded	352000	352000	0	0.0000	raw file
drop duplicate rows	352000	352000	0	0.0000	exact row duplicates
drop leakage / id columns	352000	352000	0	0.0000	id + dropoff_datetime removed (column dropped)
NYC bounding box	352000	351601	399	0.1134	$[-74.30,-73.70] \times [40.50,40.90]$
passenger_count 1–6	351601	351589	12	0.0034	drop 0 and 7+
duration 60–7200 s	351589	349080	2509	0.7136	drop cancelled / multi-day trips

Test metrics (same held-out trips for every model)

Rank	Model	Train rows	R2	RMSE	MAE	CV R2 mean	CV R2 std	Fit seconds
1	Gradient Boosting Regressor	279264	0.7961	291.4826	188.7737	0.7921	0.0028	2.2460
2	Random Forest Regressor	279264	0.7921	294.3231	189.9446	0.7860	0.0022	14.7671
3	Random Forest Regressor (tuned)	279264	0.7912	294.9744	189.7353	0.7846	0.0025	19.2083
4	Decision Tree Regressor	279264	0.7506	322.3702	211.5239	0.7451	0.0025	2.0758
5	Support Vector Regressor	12000	0.7281	336.6159	213.1370	0.7204	0.0106	2.9268
6	Polynomial Regression	279264	0.7026	352.0505	238.5700	0.6958	0.0040	1.7554
7	K-Nearest Neighbors Regressor	40000	0.6993	354.0092	243.5402	0.6854	0.0042	0.0006
8	Ridge Regression	279264	0.6241	395.7908	273.0075	0.6208	0.0039	0.0128
9	Ridge Regression (tuned)	279264	0.6241	395.7913	273.0072	0.6208	0.0039	0.0159
10	Linear Regression	279264	0.6241	395.7913	273.0072	0.6208	0.0039	0.0622
11	Lasso Regression	279264	0.6240	395.8115	273.1529	0.6206	0.0040	4.9906
12	ElasticNet Regression	279264	0.6092	403.5481	281.5870	0.6047	0.0039	0.1381
13	Mean baseline	279264	-0.0000	645.5262	462.3763	-0.0000	0.0000	0.2297

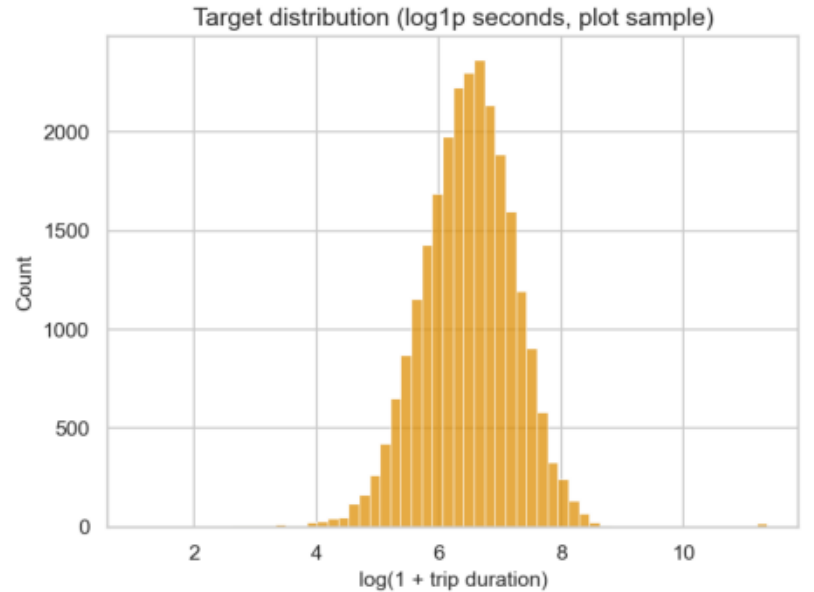
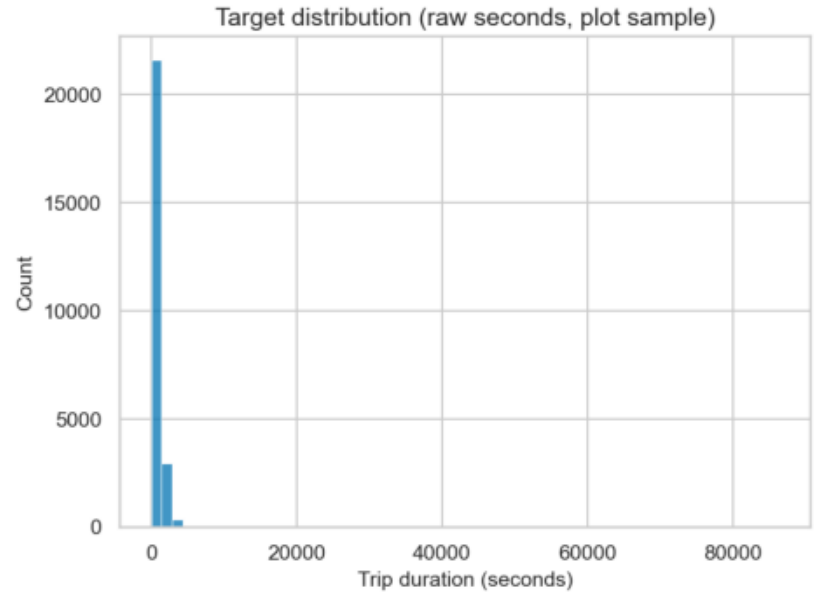
5-fold CV R² for the two best official models

Model	Test R2	Test RMSE	Test MAE	CV folds	CV R2 mean	CV R2 std	Fold scores
Gradient Boosting Regressor	0.7961084274141603	291.4825502294567	188.77367672215055	5	0.7921	0.0028	0.7948, 0.7867, 0.7938, 0.7927, 0.7948
Random Forest Regressor	0.7911940842113162	294.9743971246078	189.73529144208754	5	0.7846	0.0025	0.7868, 0.7798, 0.786, 0.7847, 0.7868

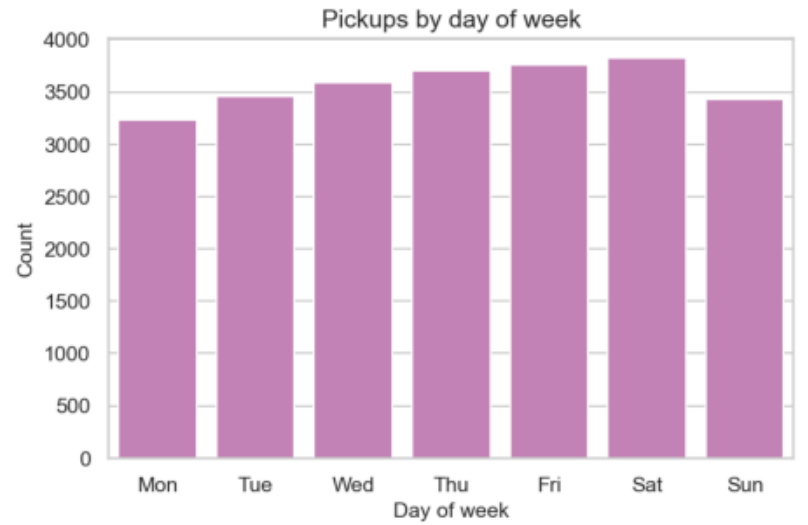
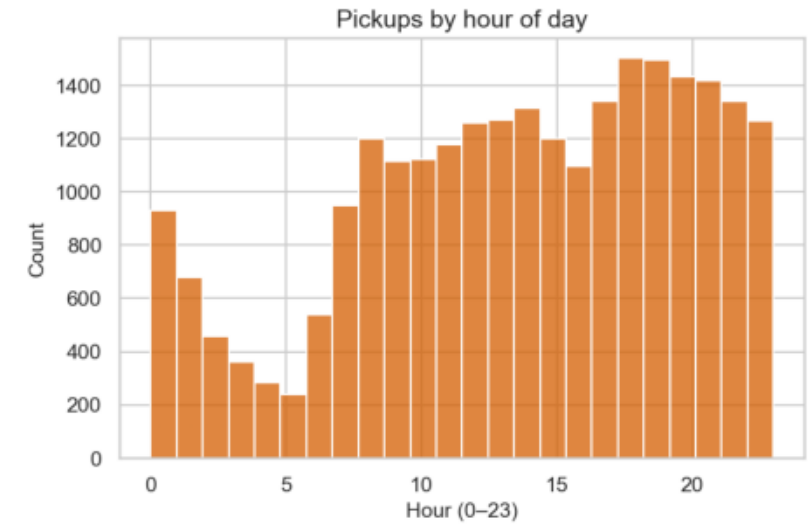
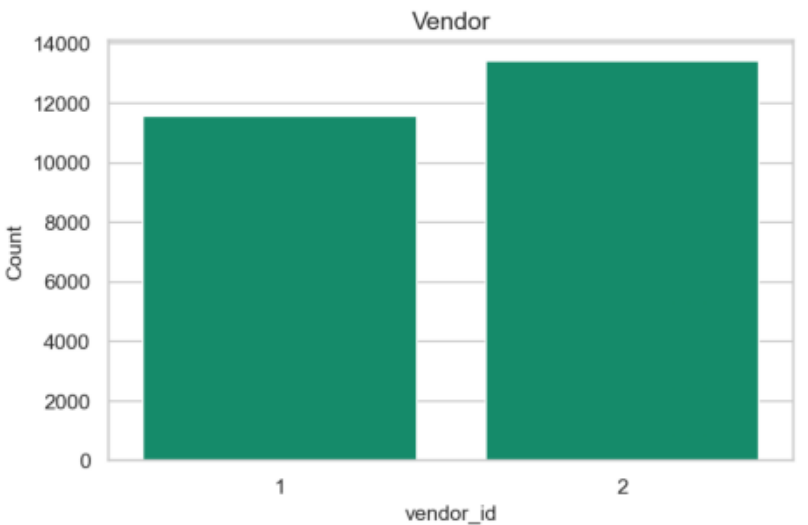
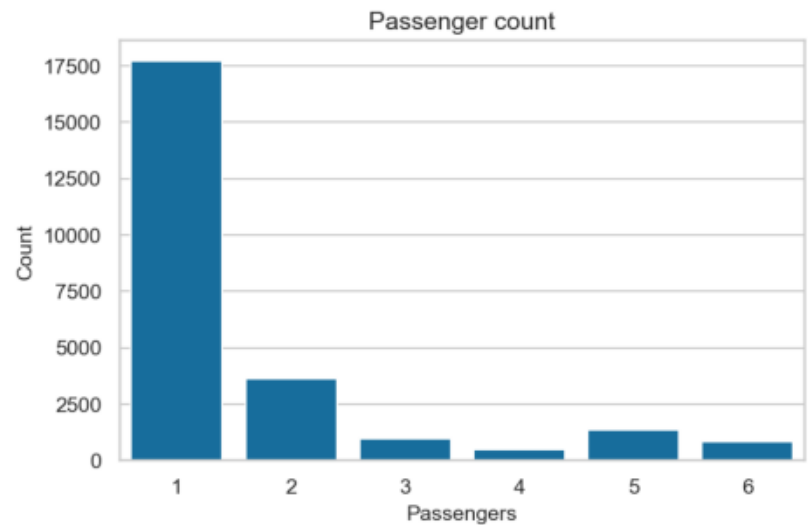
Manual tester — predicted duration for user-edited trips

trip	model_used	predicted_seconds	predicted_minutes	trip_distance_km	pickup_hour	is_rush_hour
1	Gradient Boosting Regress	675.8615	11.2644	1.5041	17	1
2	Gradient Boosting Regress	630.7121	10.5119	1.8075	0	0

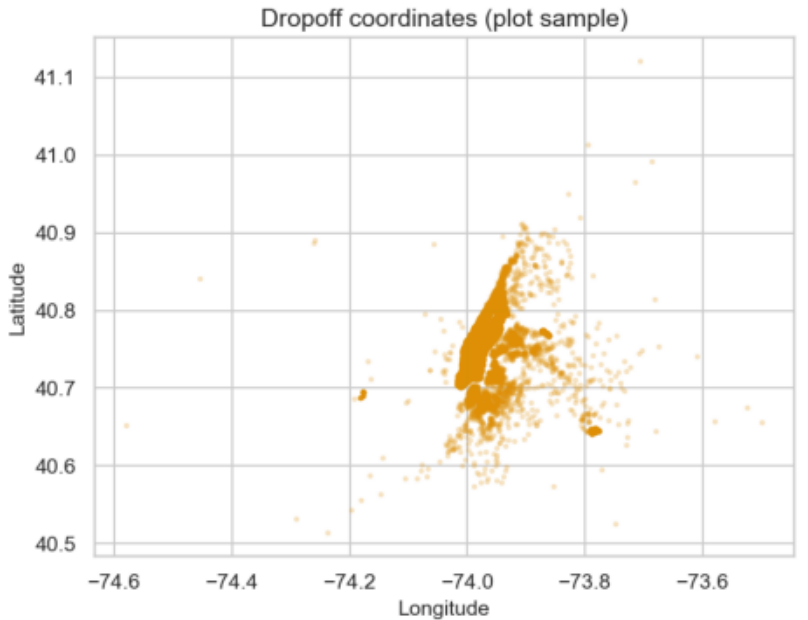
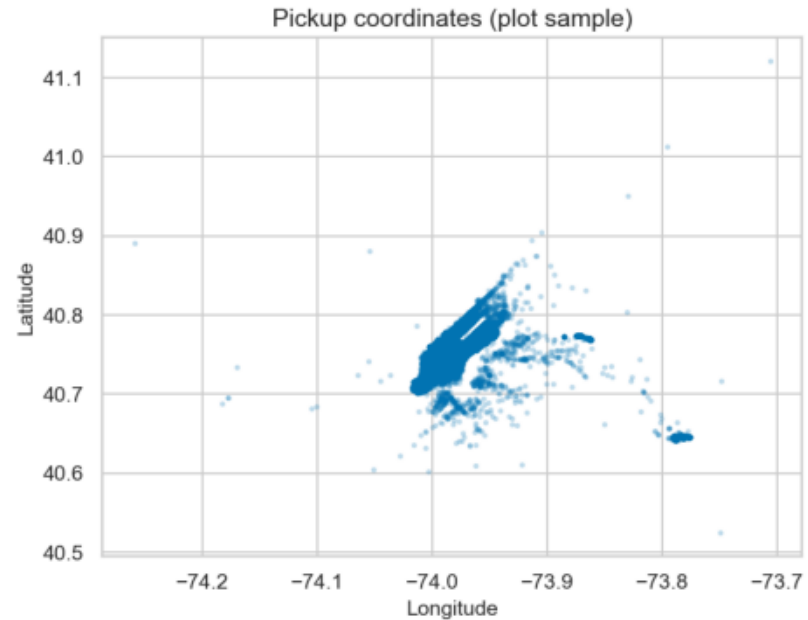
01 target distribution raw seconds plot sample



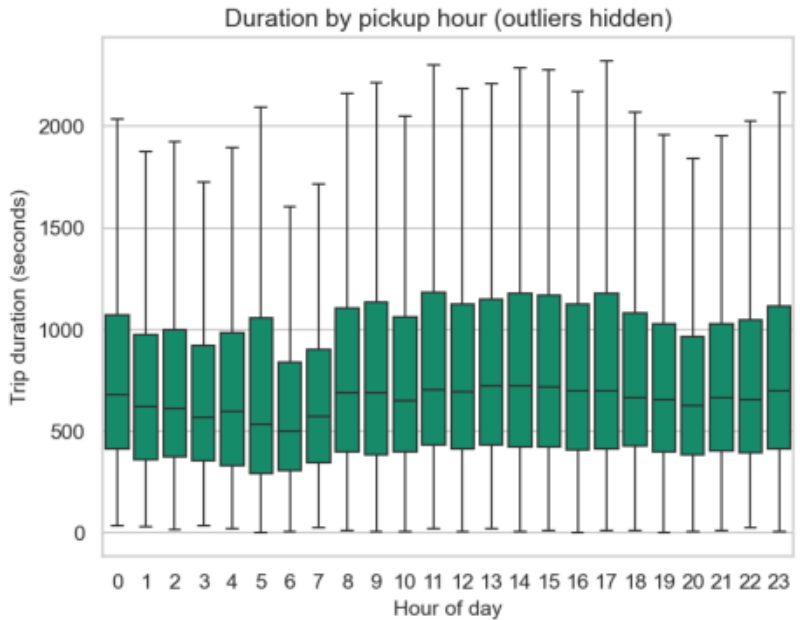
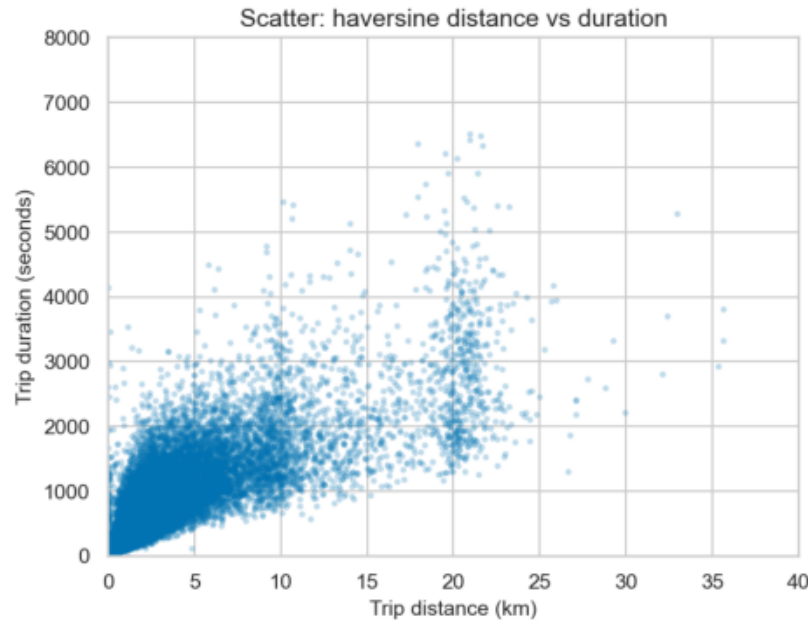
02 passenger count



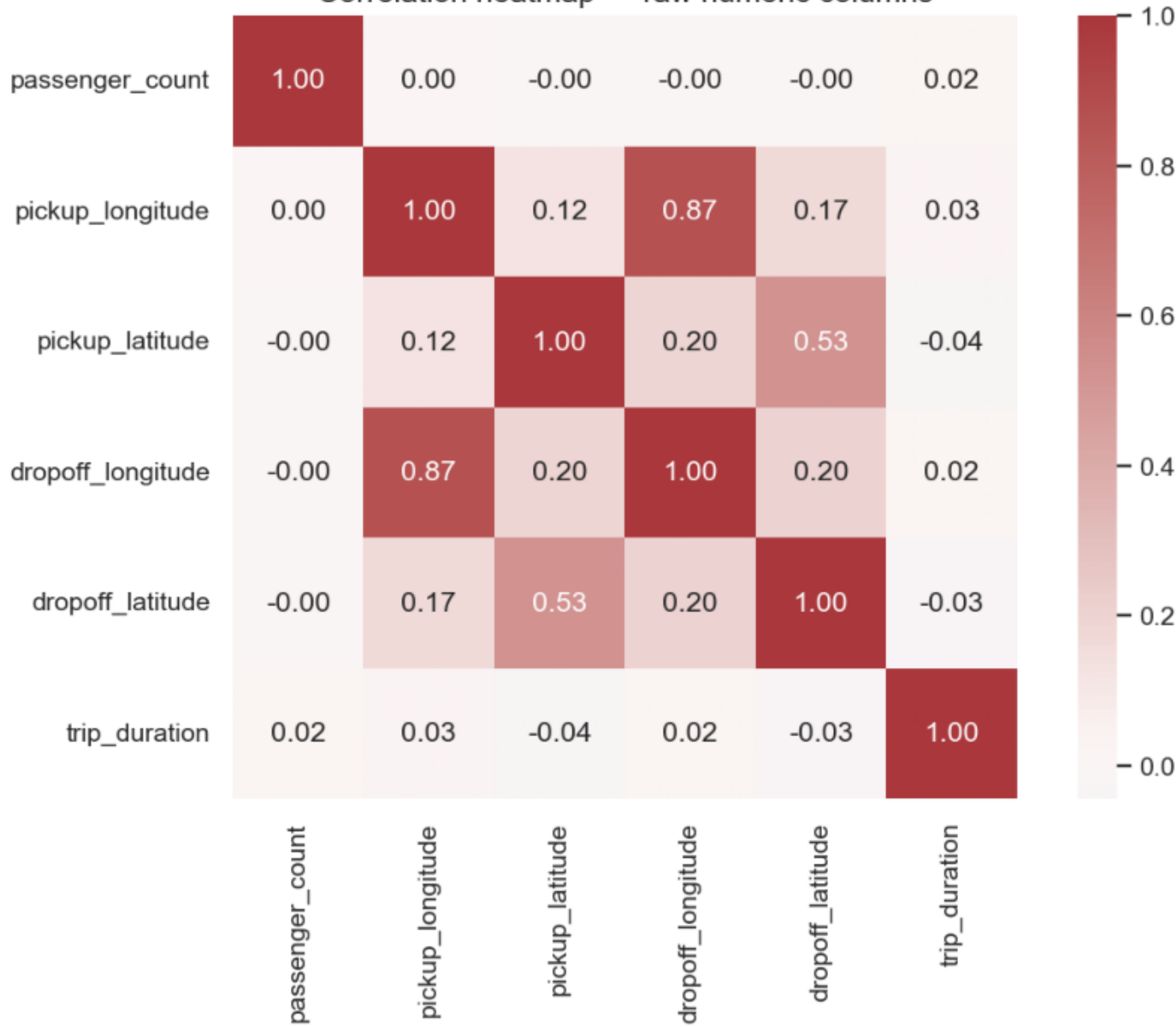
03 pickup coordinates plot sample



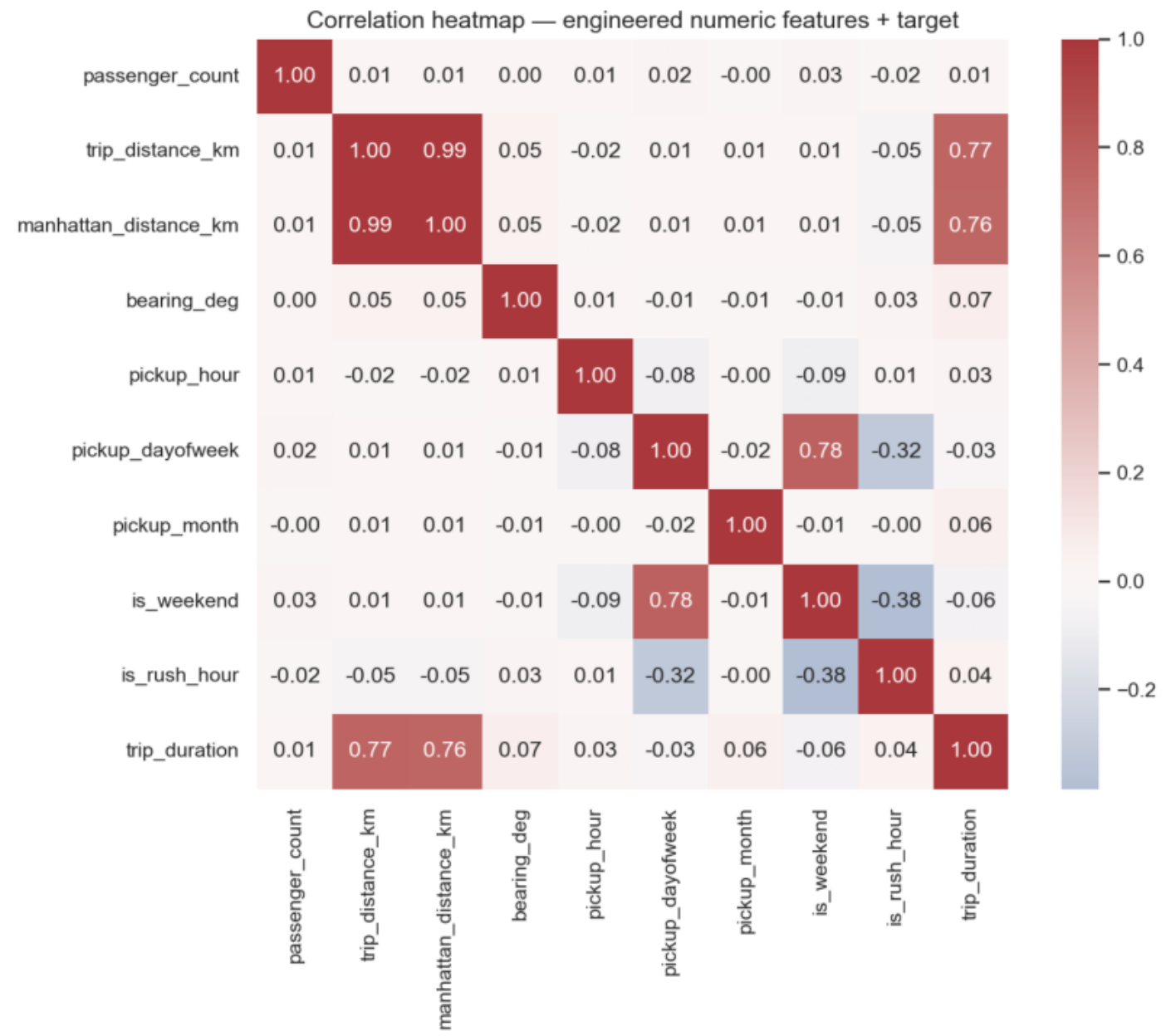
04 scatter haversine distance vs duration



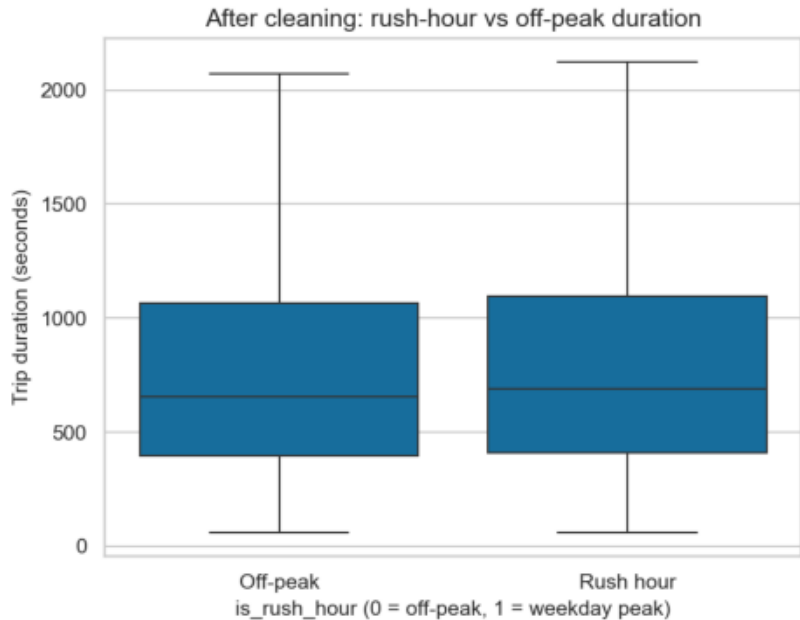
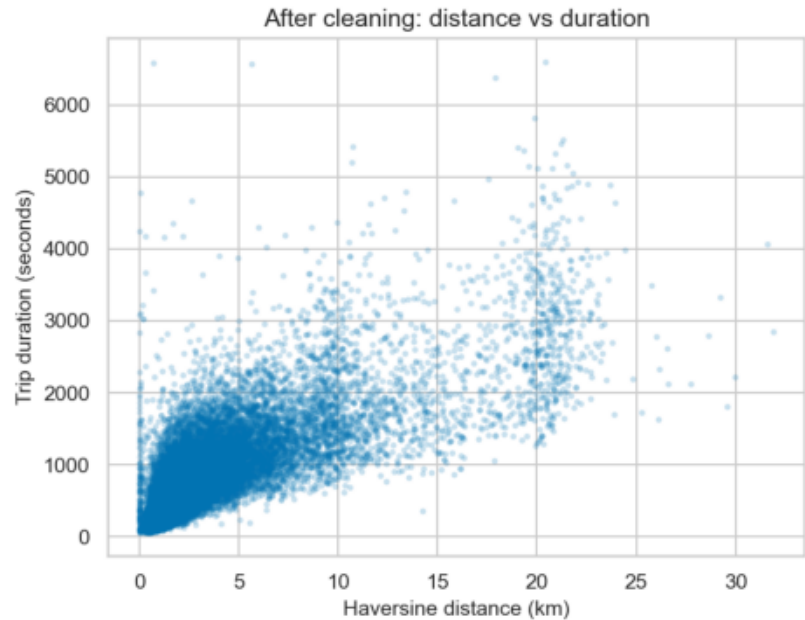
Correlation heatmap — raw numeric columns

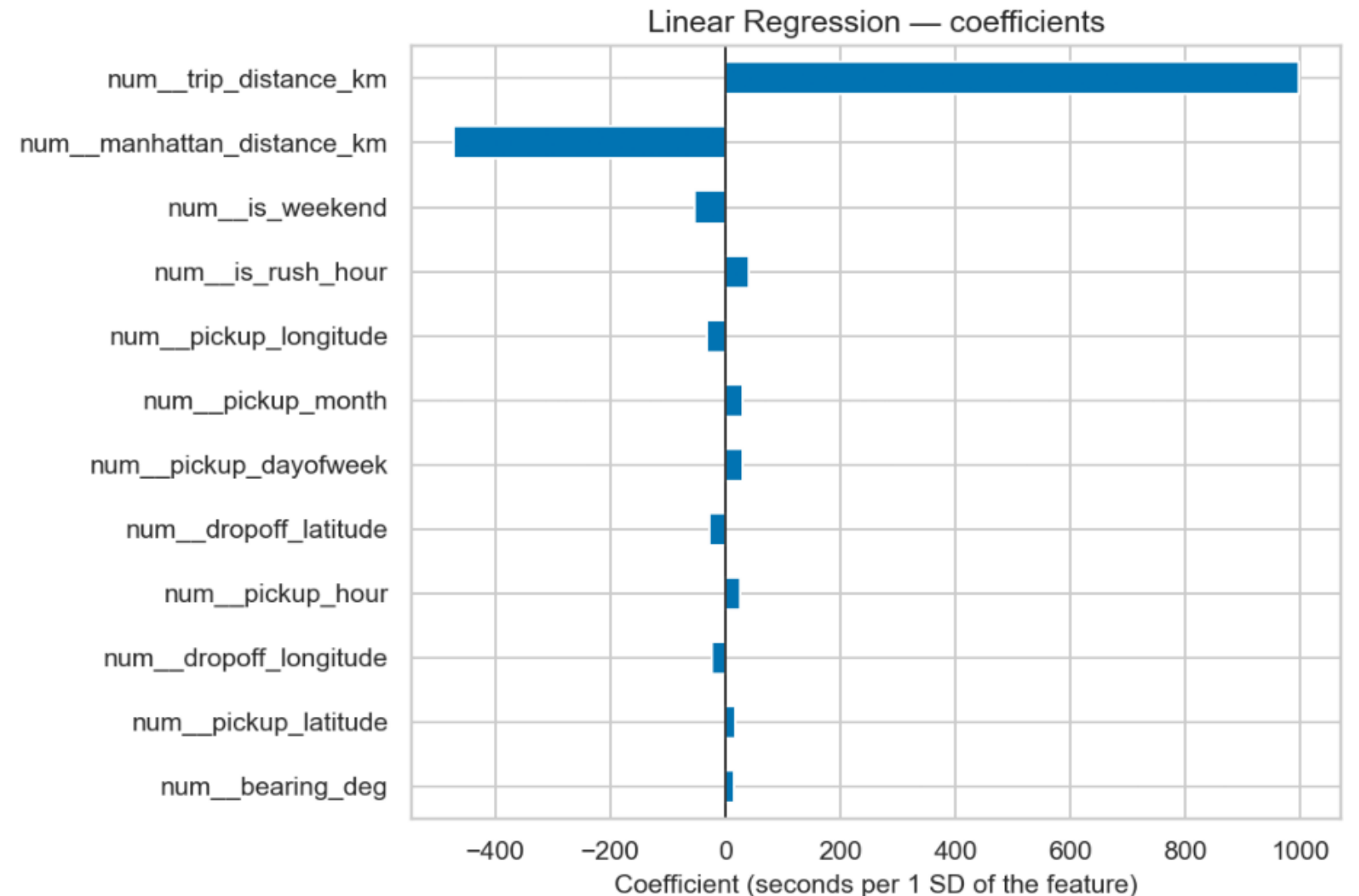


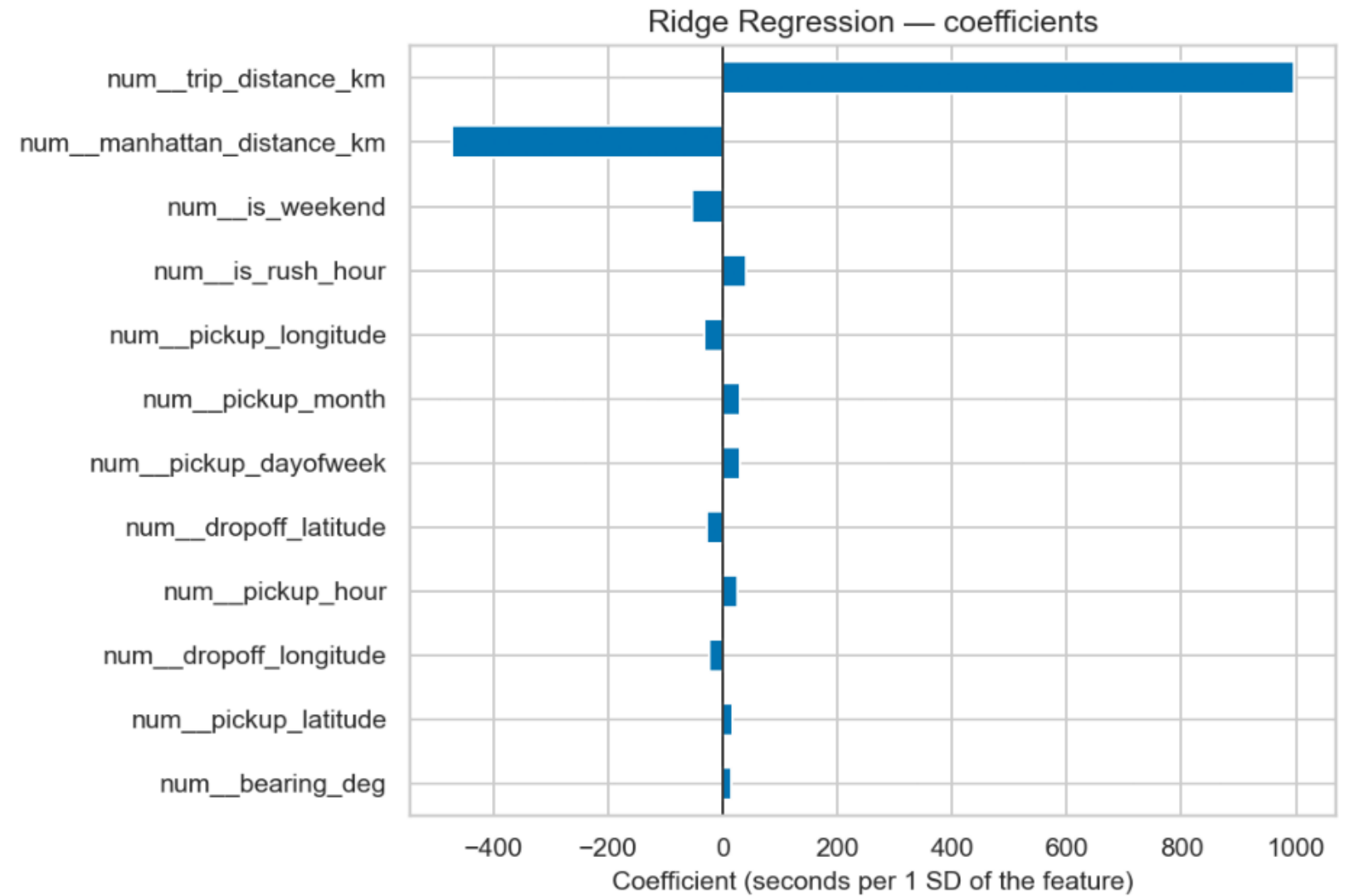
06 correlation heatmap engineered numeric features target

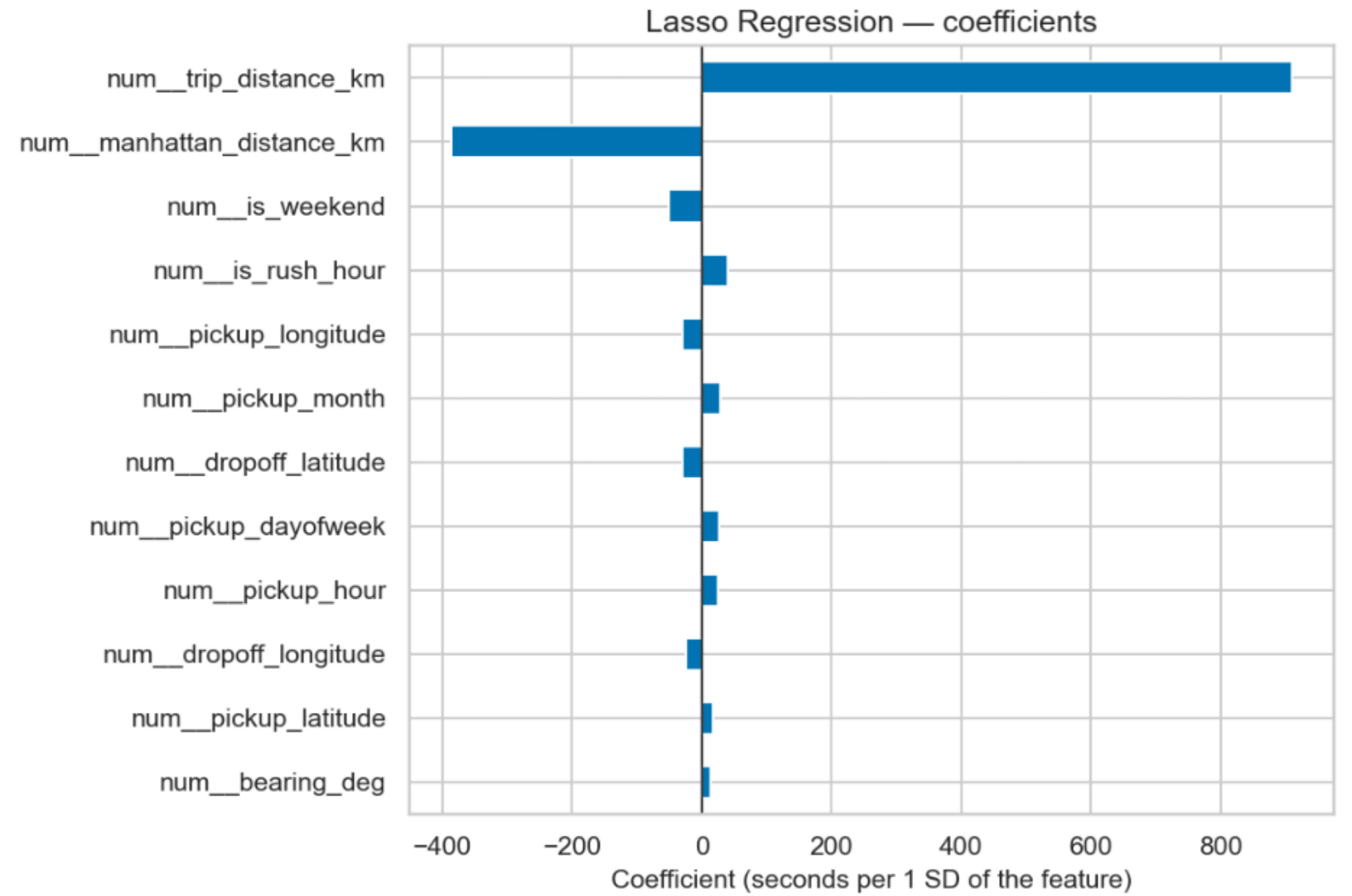


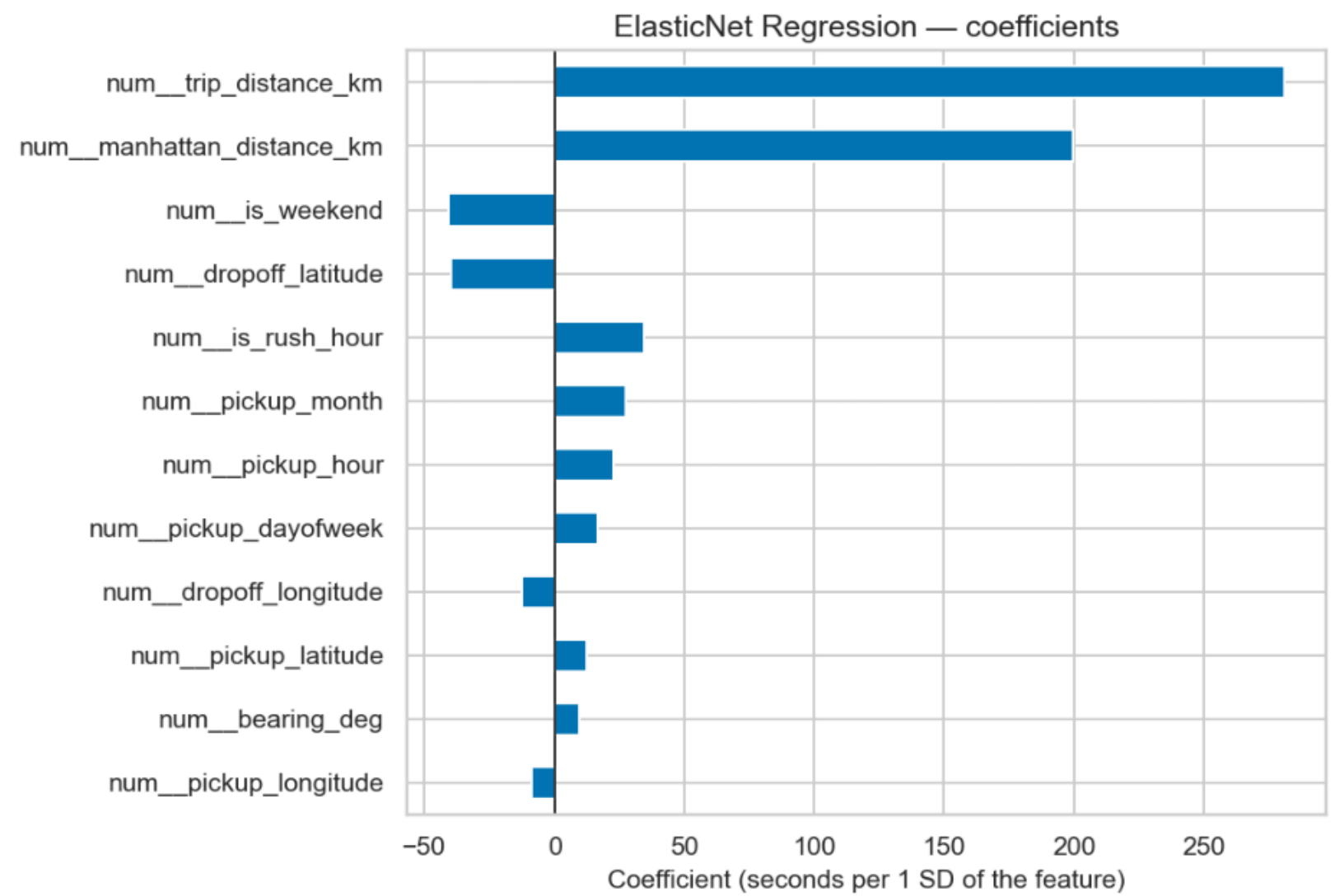
07 after cleaning distance vs duration

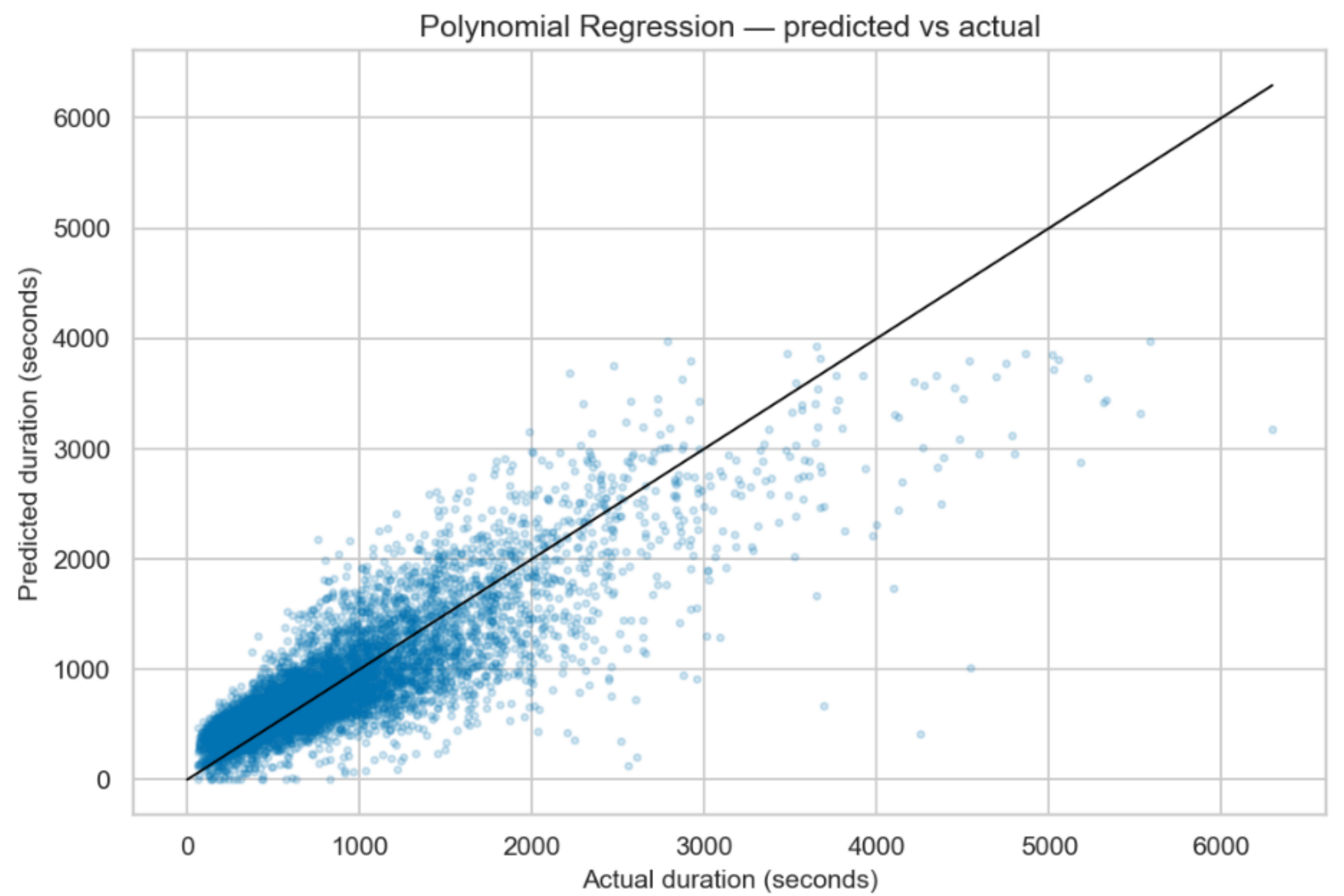


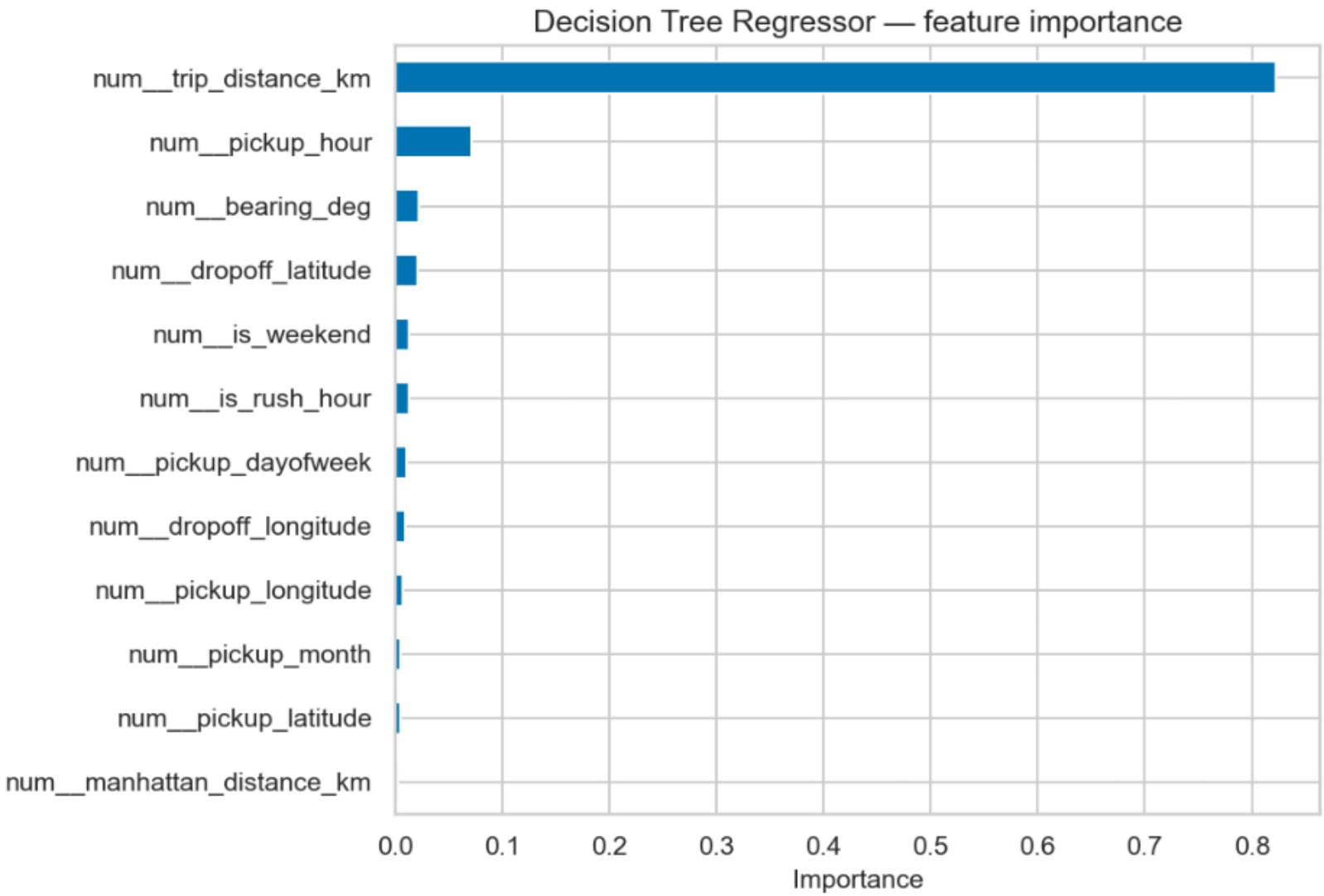


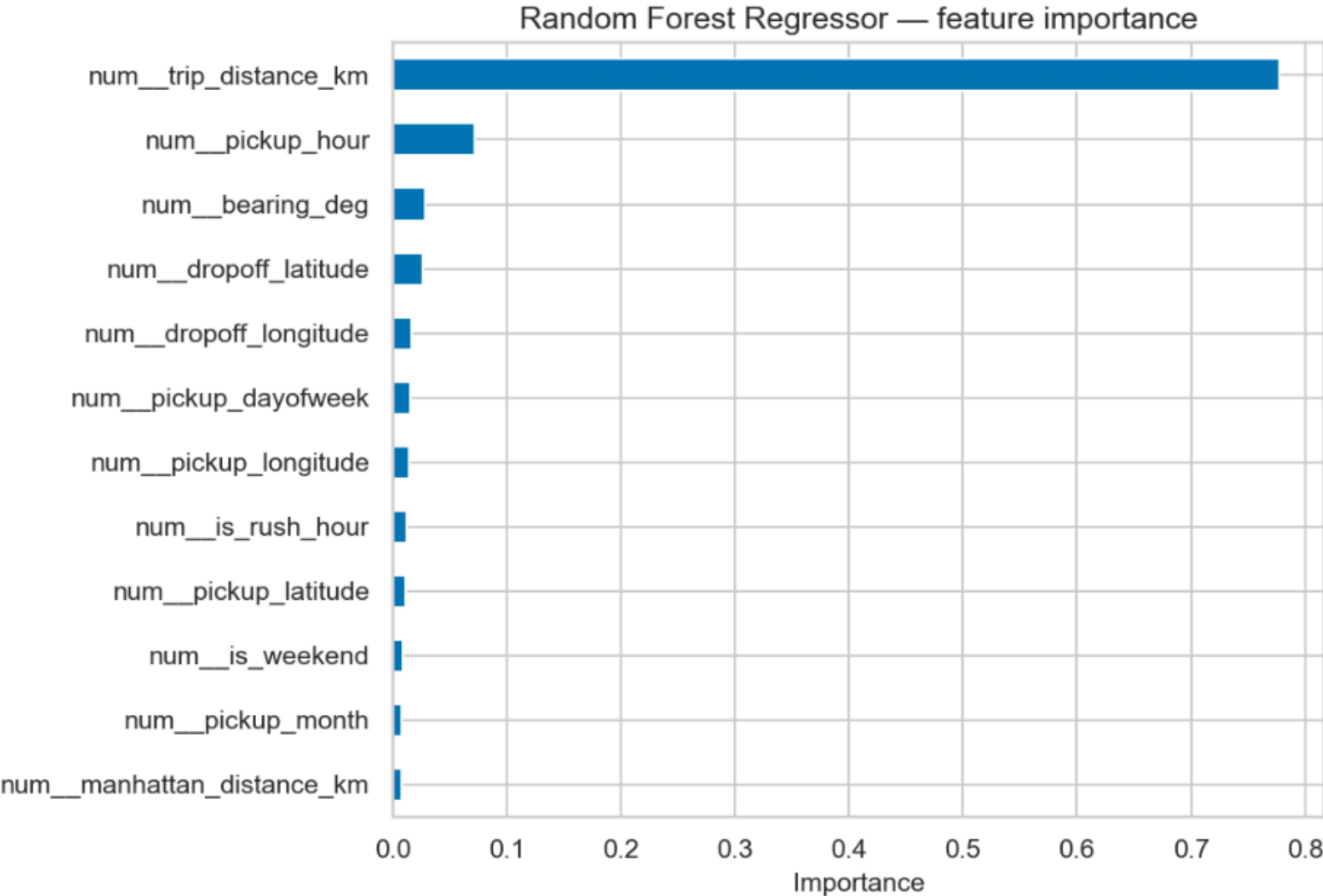


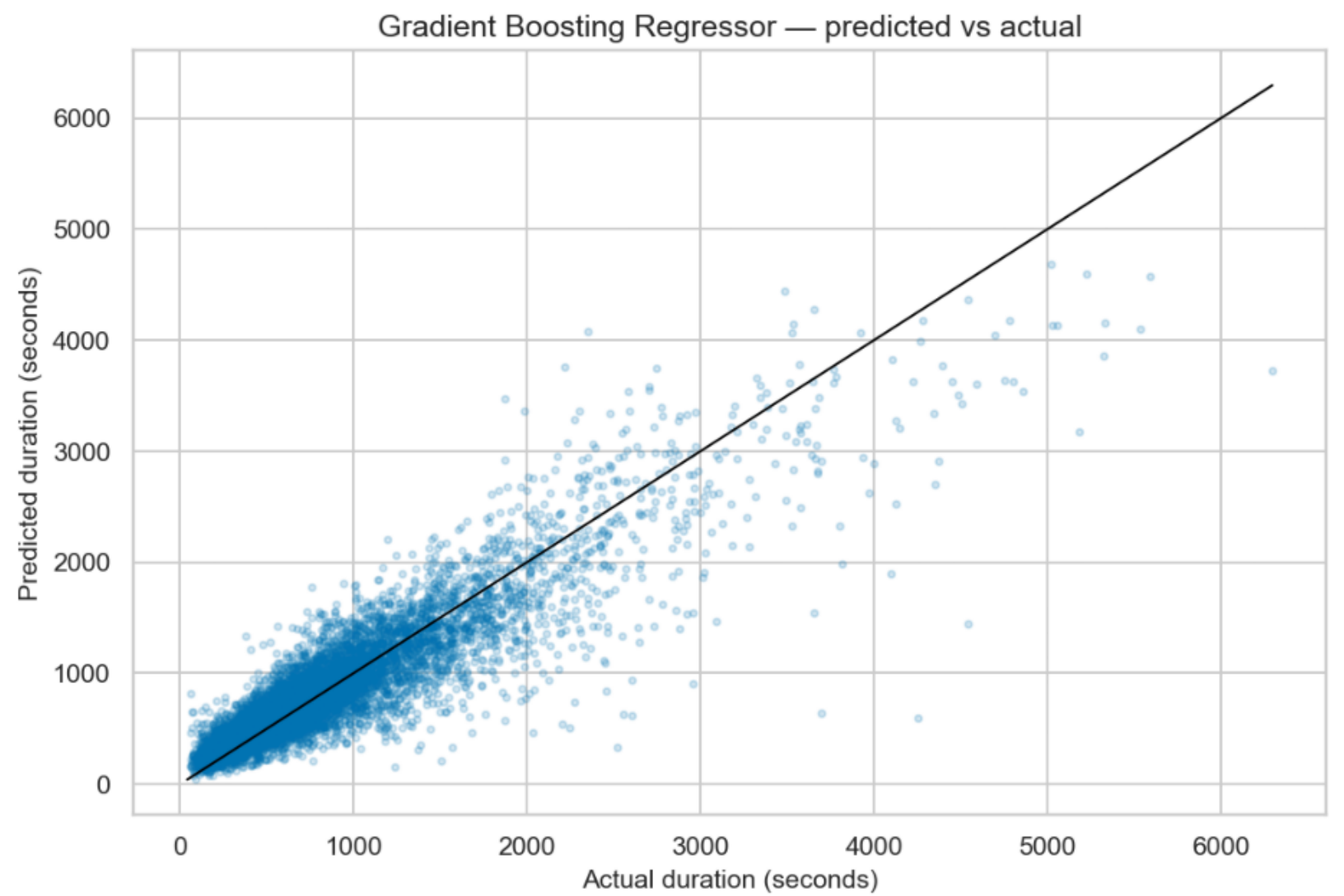


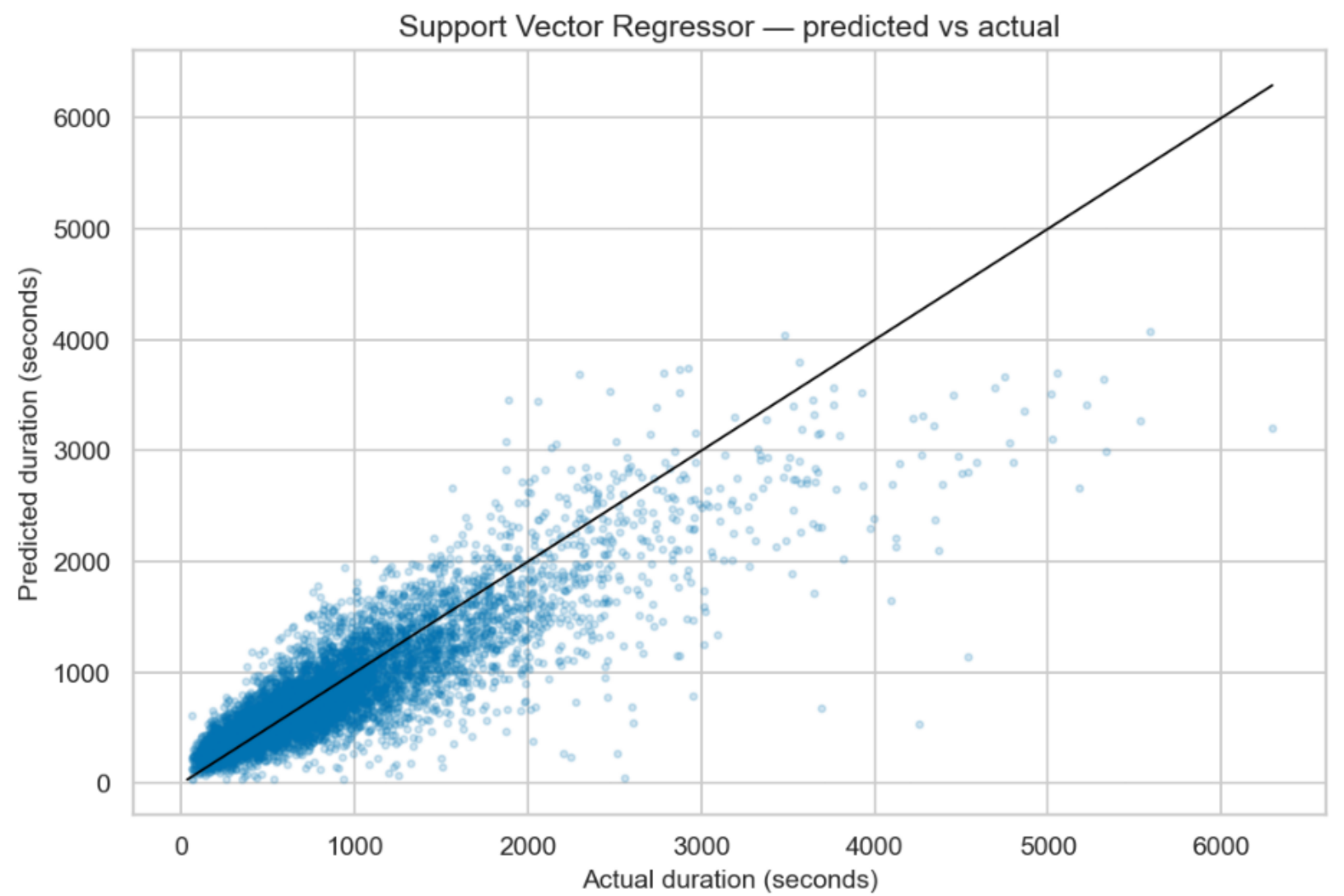


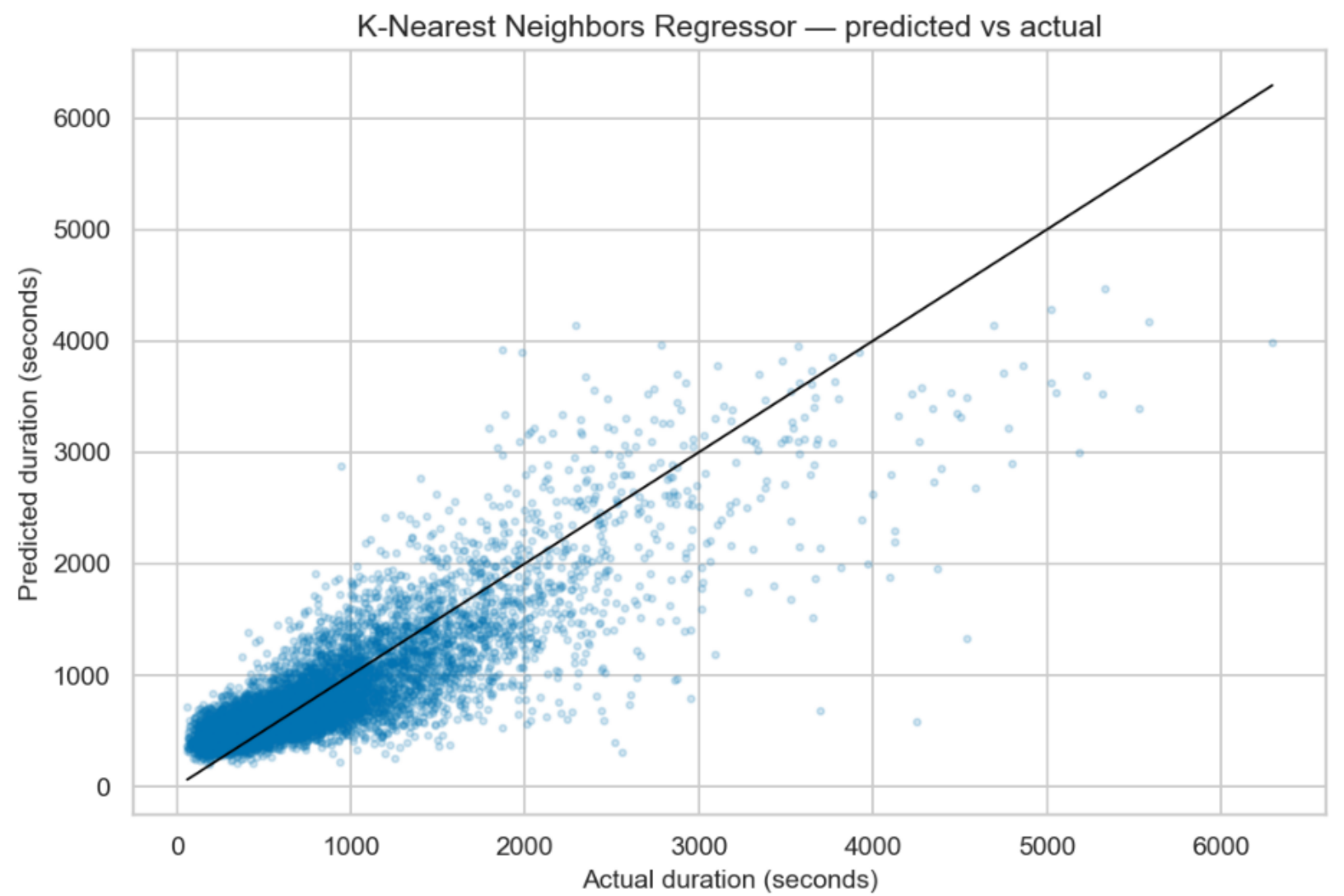




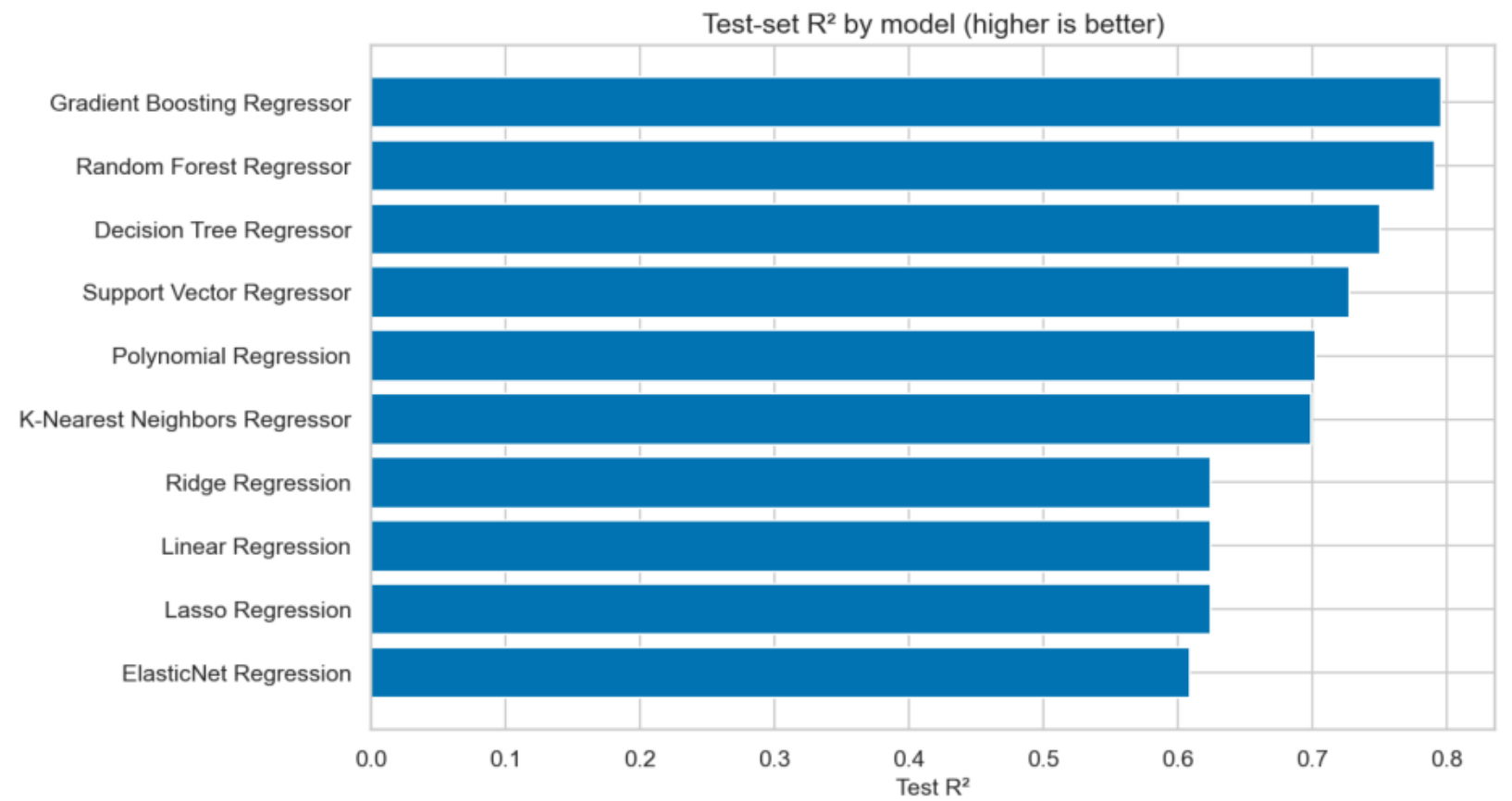


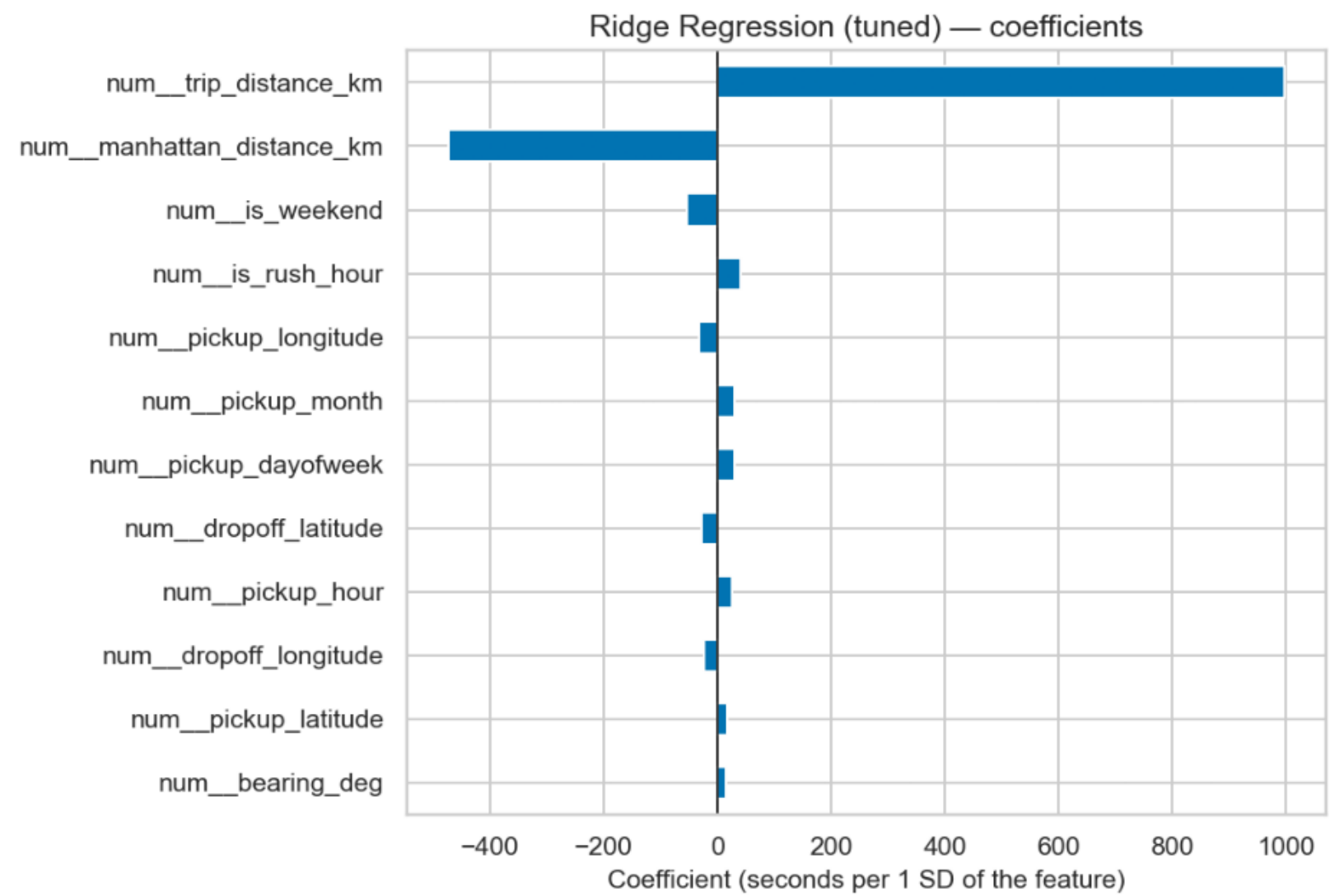


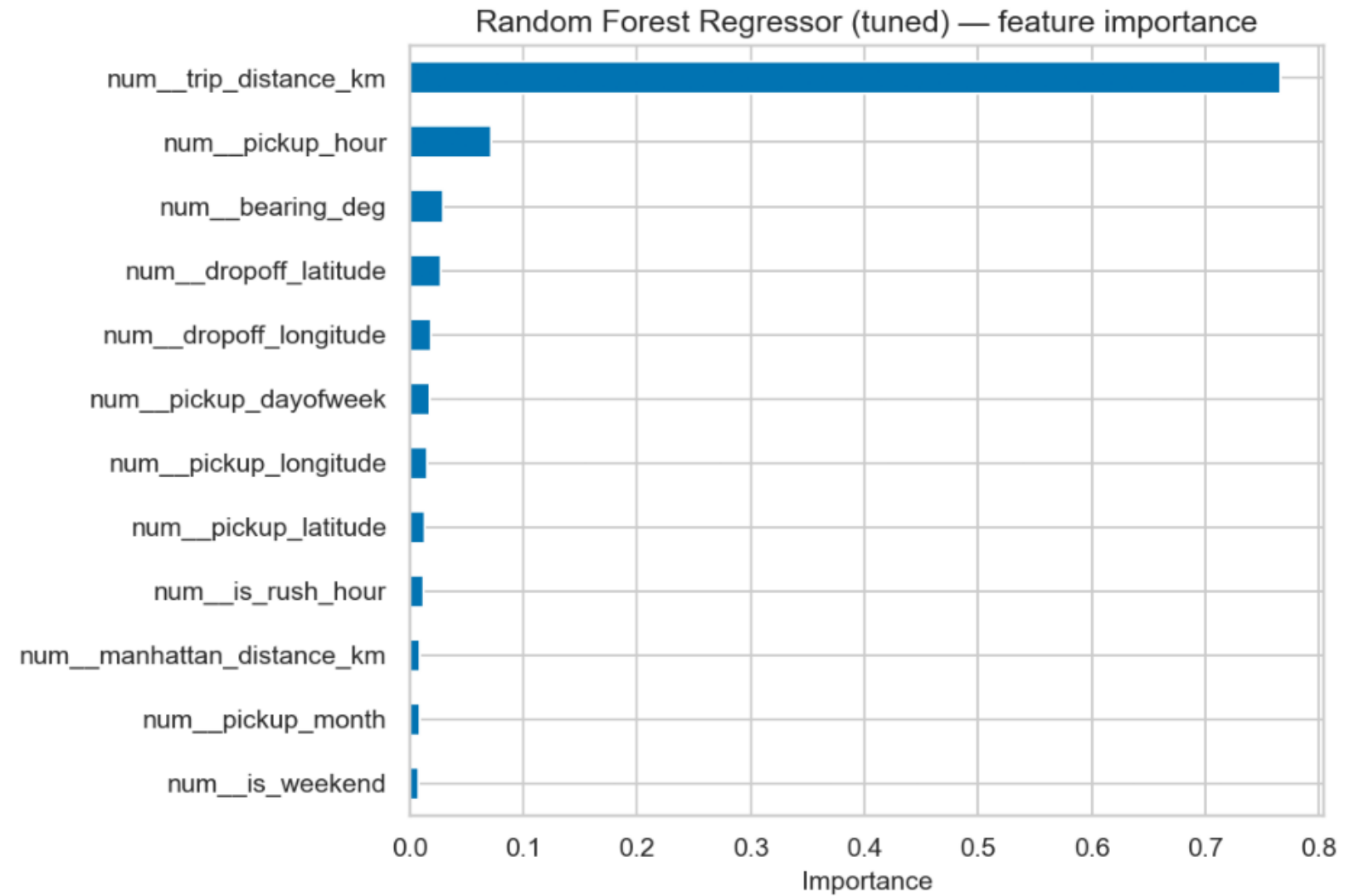




18 test set r² by model higher is better







21 predicted vs actual gradient boosting regressor

